Implied Volatility: Existence, Uniqueness, and Numerical Computation

Quant Finance Portfolio — Phase 1, Algorithm 3

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Abstract

We define the implied volatility of a European call as the unique solution to an inverse problem in the Black–Scholes model, prove that this solution exists and is unique exactly when the market price respects the model-free no-arbitrage bounds, and analyse the Newton–Raphson iteration that recovers it. The standard pathologies of Newton's method—vanishing derivative, poor initial guess—are shown to manifest in this problem in specific regimes (near expiry, deep in or out of the money, extreme volatility levels), motivating both a domain-aware initial guess (the Brenner–Subrahmanyam approximation) and a robust fallback strategy via bracketing methods.

1 The Inverse Problem

The Black–Scholes formula derived in the companion document `black_scholes.tex` expresses the no-arbitrage price of a European call option as a function

$$C^{\mathrm{BS}} : (0,\infty)^2 \times \mathbb{R} \times (0,\infty)^2 \to \mathbb{R}, \qquad (S, K, r, \sigma, T) \longmapsto C^{\mathrm{BS}}(S, K, r, \sigma, T).$$

Among the five arguments, four are observable from the market at the moment of pricing: the spot price S, the strike K, the risk-free rate r, and the time to maturity T. The fifth, the volatility σ, is a parameter of the model and is *not* directly observable. In practice, σ is inferred from market prices of options through an inverse procedure.

Definition 1.1 (Implied volatility)**.** Given observable parameters $S, K, r, T > 0$ and a market price $C^{\mathrm{mkt}} > 0$ of a European call, the *implied volatility* $\sigma^{\mathrm{IV}} = \sigma^{\mathrm{IV}}(S, K, r, T, C^{\mathrm{mkt}})$ is the value of $\sigma > 0$ satisfying

$$C^{\mathrm{BS}}(S, K, r, \sigma, T) = C^{\mathrm{mkt}}, \tag{1}$$

provided such a value exists and is unique.

The qualifier in the last clause is essential. Without it, the definition would assign no value when no σ reproduces the market price, and would be ambiguous if more than one did. We therefore begin by establishing the conditions under which (1) is a well-posed inversion problem.

Remark 1.2 (Implied volatility is a model-dependent object)*.* The implied volatility is *the volatility that, plugged into Black–Scholes, reproduces the market price.* It is not, despite the informal name, "the volatility of the underlying" in any model-independent sense. The same market price interpreted under a different model (Heston, local volatility, jump-diffusion) yields a different implied object. The Black–Scholes implied volatility is a quoting convention: a one-to-one rescaling of price into a more intuitive number with the model serving as the dictionary.

2 Existence and Uniqueness

We fix $S, K, r, T > 0$ throughout this section and study (1) as a one-dimensional equation in the unknown σ. Define the auxiliary function

$$g : (0, \infty) \to \mathbb{R}, \qquad g(\sigma) \stackrel{\text{def}}{=} C^{\text{BS}}(S, K, r, \sigma, T) - C^{\text{mkt}}.$$

The implied volatility, when it exists, is a positive root of g.

2.1 Strict monotonicity

Lemma 2.1 (Vega is strictly positive)**.** *For all $\sigma > 0$,*

$$\frac{\partial C^{\text{BS}}}{\partial \sigma}(S, K, r, \sigma, T) = S\,\varphi(d_1)\,\sqrt{T} > 0,$$

where φ denotes the standard normal density and d_1 is given by the formula in `black_scholes.tex`*.*

Proof. The closed-form expression for Vega is derived in the companion document on the Greeks. Positivity follows from $S > 0$, $\sqrt{T} > 0$, and $\varphi(x) > 0$ for every $x \in \mathbb{R}$. □

Corollary 2.2 (Strict monotonicity of g)**.** *The function $\sigma \mapsto g(\sigma)$ is strictly increasing on $(0, \infty)$. In particular, if a root of g exists, it is unique.*

Proof. g and $C^{\text{BS}}(\,\cdot\,)$ differ by a constant, so g inherits the strict positivity of the derivative from Lemma 2.1. A strictly increasing function takes each value in its image exactly once. □

2.2 Asymptotic behaviour

To prove existence we examine the limits of C^{BS} at the boundaries of $\sigma > 0$.

Proposition 2.3 (Limits of C^{BS} in σ)**.** *Fix $S, K, r, T > 0$. Then*

$$\lim_{\sigma \to 0^+} C^{\text{BS}}(S, K, r, \sigma, T) = \left(S - Ke^{-rT}\right)^+, \tag{2}$$

$$\lim_{\sigma \to \infty} C^{\text{BS}}(S, K, r, \sigma, T) = S. \tag{3}$$

Proof. **Limit $\sigma \to 0^+$.** The argument d_1 of the BS formula is

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = \frac{\log(S/K) + rT}{\sigma\sqrt{T}} + \tfrac{1}{2}\sigma\sqrt{T}.$$

The second term vanishes as $\sigma \to 0^+$. The first term diverges to $+\infty$, $-\infty$, or is undefined depending on the sign of $\log(S/K) + rT = \log(S/(Ke^{-rT}))$:

- If $S > Ke^{-rT}$ (forward-ITM), then $d_1, d_2 \to +\infty$, so $N(d_1), N(d_2) \to 1$, and $C^{\text{BS}} \to S - Ke^{-rT} = (S - Ke^{-rT})^+$.
- If $S < Ke^{-rT}$ (forward-OTM), then $d_1, d_2 \to -\infty$, so $N(d_1), N(d_2) \to 0$, and $C^{\text{BS}} \to 0 = (S - Ke^{-rT})^+$.
- If $S = Ke^{-rT}$, then the leading term in d_1, d_2 is $\pm\frac{1}{2}\sigma\sqrt{T} \to 0$, so $N(d_1), N(d_2) \to \frac{1}{2}$, and $C^{\text{BS}} \to \frac{1}{2}(S - Ke^{-rT}) = 0 = (S - Ke^{-rT})^+$.

All three branches agree with (2).

Limit $\sigma \to \infty$. As $\sigma \to \infty$, the $\frac{1}{2}\sigma^2 T$ term dominates the numerator of d_1, so $d_1 \to +\infty$ and $d_2 = d_1 - \sigma\sqrt{T} \to -\infty$ (since $\sigma\sqrt{T}$ grows faster than d_1). Hence $N(d_1) \to 1$, $N(d_2) \to 0$, and $C^{\text{BS}} \to S$. □

2.3 Main result

Combining strict monotonicity with the boundary behaviour gives a sharp characterisation of when implied volatility is well-defined.

Theorem 2.4 (Existence and uniqueness of implied volatility)**.** *Fix $S, K, r, T > 0$ and $C^{\mathrm{mkt}} \in \mathbb{R}$. The equation* (1) *has a unique solution $\sigma^{\mathrm{IV}} \in (0, \infty)$ if and only if*

$$\left(S - Ke^{-rT}\right)^+ < C^{\mathrm{mkt}} < S. \tag{4}$$

Proof. By Corollary 2.2, C^{BS} is strictly increasing in σ on $(0, \infty)$. By Proposition 2.3 and continuity, the image of $(0, \infty)$ under $\sigma \mapsto C^{\mathrm{BS}}$ is the open interval $\big((S - Ke^{-rT})^+, S\big)$. Hence (1) has a solution if and only if C^{mkt} lies in this open interval, and when it does, the solution is unique by strict monotonicity. □

Remark 2.5 (Connection with no-arbitrage)*.* The bounds in (4) are not artefacts of the Black–Scholes model. They are the model-free no-arbitrage bounds for a European call: on any market admitting no arbitrage, every quoted call price must satisfy them, regardless of which model one believes generates returns. Theorem 2.4 can therefore be restated as: *within the Black–Scholes framework, every no-arbitrage price admits a unique implied volatility, and conversely, prices outside the no-arbitrage range cannot be reproduced by any choice of σ.*

This duality is conceptually important. A pricing routine that fails to find an implied volatility is signalling either (i) corrupt input data, (ii) mismatched parameters (e.g. wrong rate or wrong spot), or (iii) a genuine arbitrage opportunity. The first two are common; the third is rare but real. In all cases, the failure is informative and should be surfaced to the user, not silently absorbed.

Remark 2.6 (Behaviour in the second derivative)*.* A direct calculation gives

$$\frac{\partial^2 C^{\mathrm{BS}}}{\partial \sigma^2} = \frac{S\,\varphi(d_1)\,\sqrt{T}}{\sigma}\, d_1\, d_2.$$

Unlike the first derivative, the sign of the second derivative is not constant: it agrees with the sign of $d_1 d_2$. For options near the forward-ATM region with moderate volatility, d_1 and d_2 typically have opposite signs and C^{BS} is locally concave in σ; for options far from ATM, or for large σ, the two agree in sign and C^{BS} is convex. The change of curvature occurs at the locus $d_1 d_2 = 0$. This non-uniform convexity has direct consequences for Newton's method, as we discuss below.

3 Newton–Raphson for Implied Volatility

Theorem 2.4 establishes the well-posedness of the inversion. We turn now to its computation.

3.1 The iteration

The Newton–Raphson iteration applied to $g(\sigma) = 0$ is

$$\sigma_{n+1} = \sigma_n - \frac{g(\sigma_n)}{g'(\sigma_n)} = \sigma_n - \frac{C^{\mathrm{BS}}(\sigma_n) - C^{\mathrm{mkt}}}{\mathcal{V}(\sigma_n)}, \tag{5}$$

where $\mathcal{V}(\sigma) = S\,\varphi(d_1)\,\sqrt{T}$ is Vega evaluated at σ.

The iteration is *cheap*: each step requires one evaluation of C^{BS} and one of $\mathcal{V}$, both already implemented. The iteration is *geometrically natural*: the next iterate is the intersection of the tangent line to g at $(\sigma_n, g(\sigma_n))$ with the horizontal axis.

Proposition 3.1 (Local quadratic convergence)**.** *Suppose $\sigma^* > 0$ is the unique root of g, and let σ_n be the n-th Newton iterate. There exists a neighbourhood U of σ^* such that, if $\sigma_0 \in U$, the sequence σ_n converges to σ^* with*

$$|\sigma_{n+1} - \sigma^*| \leq C\,|\sigma_n - \sigma^*|^2$$

for some constant $C > 0$ depending on $g'(\sigma^*)$ and on the sup-norm of g'' on U.

The proof is standard (Taylor expansion of g around σ^* combined with $g'(\sigma^*) > 0$) and is omitted; see, e.g., Burden and Faires, *Numerical Analysis*, Theorem 2.9.

3.2 Failure modes

The proposition is local: it guarantees fast convergence *once σ_n is close enough.* The behaviour from arbitrary starting points can fail in three ways specific to the implied volatility problem.

Vanishing Vega. The Newton step in (5) is inversely proportional to $\mathcal{V}(\sigma_n)$. When Vega is small, the step becomes arbitrarily large even for moderate values of the residual $g(\sigma_n)$, and the next iterate is propelled into a region of parameter space unrelated to the root.

Vega vanishes (as a function of σ) in two distinct regimes:

- *Near expiry.* As $T \to 0$, $\mathcal{V} = S\varphi(d_1)\sqrt{T} \to 0$ pointwise in σ, with the exception of the ATM case where a more careful expansion shows $\mathcal{V} \to 0$ at the slower rate $O(\sqrt{T})$.
- *Deep in or out of the money.* For $|\log(S/K)| \gg \sigma\sqrt{T}$, one has $d_1 \to \pm\infty$, so $\varphi(d_1) \to 0$ super-exponentially.

In both regimes Newton becomes unsafe. A practical implementation must detect $\mathcal{V}(\sigma_n) < \varepsilon$ for some threshold $\varepsilon > 0$ and abandon Newton in favour of a bracketing method (see Section 4).

Domain violation. The Newton update (5) can produce $\sigma_{n+1} \leq 0$. The iteration is unaware of the constraint $\sigma > 0$; a single overlong step can push the iterate out of the feasible domain, after which $g(\sigma_{n+1})$ is undefined. Detection is straightforward (test the sign of σ_{n+1} before proceeding) but the response must be a fallback, not a retry.

Curvature flip. By Remark 2.6, the second derivative of C^{BS} in σ changes sign across the locus $d_1 d_2 = 0$. When the initial guess σ_0 and the root σ^* lie on opposite sides of this locus, Newton can overshoot and oscillate. While such oscillation is rare in practice for sensible initial guesses, it is a reminder that monotonicity of g is not by itself enough to guarantee global convergence of Newton's method—convexity (or a proxy for it, like a good initial guess) is also needed.

3.3 Initial guess: Brenner–Subrahmanyam

The performance of Newton depends sensitively on σ_0. A constant guess (such as $\sigma_0 = 0.2$) works in most ordinary cases but fails for markets with extreme implied volatilities (very short-dated equity options, low-rate bond options, distressed assets). A data-driven initial guess is preferable.

The classical proposal is due to Brenner and Subrahmanyam (1988):

Proposition 3.2 (Brenner–Subrahmanyam initial guess)**.** *For an at-the-money call ($S = K$), the leading-order asymptotic behaviour of C^{BS} for small $\sigma\sqrt{T}$ is*

$$C^{\mathrm{BS}}_{\mathrm{ATM}} \;\approx\; S\,\sigma\sqrt{\frac{T}{2\pi}}, \tag{6}$$

yielding the inverted approximation

$$\sigma_0 \;=\; \sqrt{\frac{2\pi}{T}}\,\frac{C^{\mathrm{mkt}}}{S}. \tag{7}$$

For options moderately away from the money, the corresponding approximation is

$$\sigma_0 \;=\; \sqrt{\frac{2\pi}{T}}\,\frac{C^{\mathrm{mkt}} - \frac{1}{2}(S-K)}{S}. \tag{8}$$

Sketch. Setting $S = K$ in the Black–Scholes formula and Taylor-expanding $N(d_1) - N(d_2)$ around $\sigma\sqrt{T} = 0$,

$$N(d_1) - N(d_2) = \int_{-\sigma\sqrt{T}/2}^{\sigma\sqrt{T}/2} \varphi(u)\,du = \sigma\sqrt{T}\,\varphi(0) + O\big((\sigma\sqrt{T})^3\big) = \frac{\sigma\sqrt{T}}{\sqrt{2\pi}} + O\big((\sigma\sqrt{T})^3\big).$$

At ATM, $C^{\mathrm{BS}} = S(N(d_1) - N(d_2)) + O(rT)$, and to leading order this gives (6). The correction term in (8) adjusts for moneyness by subtracting half of the intrinsic value, which is the linear-in-K approximation to the deviation of C^{BS} from its ATM value. See the original paper for the full derivation. □

Remark 3.3 (Quality of the approximation). The Brenner–Subrahmanyam guess is quantitatively excellent in the near-ATM regime ($|\log(S/K)| \lesssim 0.3$) and degrades gracefully outside it. In all but the most exotic regimes it places σ_0 within the basin of quadratic convergence of Proposition 3.1, reducing the typical Newton iteration count to between three and six.

4 Robust Computation: Fallback to Bracketing

A robust implementation of implied volatility cannot rely on Newton alone. Three failure modes have been identified (vanishing Vega, domain violation, slow convergence after a fixed iteration budget); each calls for the same response: abandon Newton, fall back to a method that is guaranteed to converge.

4.1 Bracketing methods

A *bracketing method* for finding a root of g maintains at each step an interval $[a_n, b_n]$ with $g(a_n)\,g(b_n) < 0$, so that by continuity a root is contained inside. The simplest bracketing method, *bisection*, halves the interval at every step, achieving linear convergence with rate 1/2. Bisection is unconditionally convergent but slow: roughly 50 iterations are required for fifteen-digit accuracy.

For implied volatility, an initial bracket is always available. By Theorem 2.4, the root lies in some bounded positive interval; the conservative choice $[a, b] = [10^{-4}, 10]$ suffices for any realistic market data. The lower endpoint avoids the domain singularity at $\sigma = 0$; the upper endpoint accommodates extreme volatility regimes.

4.2 Hybrid algorithms

Brent's algorithm (Brent, 1973) and the related TOMS-748 algorithm (Alefeld, Potra, Shi, 1995) combine bracketing with superlinear methods (secant interpolation, inverse quadratic interpolation) to achieve unconditional convergence at superlinear rate on smooth problems. The algorithmic details are intricate but well-documented in standard references; we adopt them as a black box, deferring to existing high-quality implementations:

- In Python, `scipy.optimize.brentq` provides a production-grade implementation of Brent's algorithm.
- In C++, the standard library does not include a root-finder. We will use a direct bisection implementation as fallback, accepting the loss of speed in exchange for self-contained code with no external dependencies.

The complete computational strategy can therefore be summarised as follows.

1. Verify the no-arbitrage bounds (4). If C^{mkt} violates them, return an error.
2. Initialise σ_0 via the Brenner–Subrahmanyam formula (8), clamped to a safe positive range.
3. Iterate Newton–Raphson (5), with safeguards that detect (i) vanishing Vega, (ii) negative iterates, and (iii) excessive iteration count.
4. Upon any of the safeguard triggers, fall back to a bracketing method on the conservative interval $[10^{-4}, 10]$.

This structure is standard in production implied-volatility code, e.g., as found in the QuantLib library.

5 Conceptual Remarks

Remark 5.1 (The smile and the model). For a fixed underlying, Theorem 2.4 associates one implied volatility per market price. In practice, every traded strike and every traded maturity yields its own implied volatility. If the Black–Scholes model held, all of these would coincide: a single σ would reproduce every price simultaneously. Empirically they do not. The function $(K, T) \mapsto \sigma^{\mathrm{IV}}(K, T)$ is called the *implied volatility surface*; its non-flatness is the clearest empirical refutation of the Black–Scholes model and the starting point for every more sophisticated framework (local volatility, stochastic volatility, jump-diffusion). The implied volatility function is therefore best understood not as "the" model parameter but as a *quoting language*: a coordinate change from prices to numbers traders can compare across strikes and maturities.

Remark 5.2 (Implied volatility as a language). The replacement of price quotes by implied-volatility quotes is the single most consequential application of the Black–Scholes model that does *not* rely on the model being literally true. Even for desks that price options under fundamentally different dynamics, the implied volatility extracted via (1) remains the universal communication primitive. The mathematical content of this report is, in this sense, infrastructure: it ensures that the quoting language is well-defined, computable, and free of internal inconsistencies.